

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

HONEST HEADLINE AND SCOPE

Confirmed narrowly: trained independent models share mainly local structure that linear maps make useful across modalities and lineages. **Bounded:** not global or universal; scale is not monotonic; natural-world data matters; and visible geometry need not transfer.

RESEARCH PROGRAM

8-encoder main roster · 14 encoders touched · 2 shipped systems

Experiment A	Experiment B	Cheap Adapter Hub
GPT-2 vs Pythia-160M	MobileCLIP → SigLIP 2	7 encoders build it; ConvNeXt tests it
CKA + layer-wise ridge on 10,000 passages.	512×768 ridge matrix, 0.79 MB fp16 Core ML.	DINOv2 s/b/l + GPT-2 + BERT + SBERT + bge-m3 → 512-d whitened-PCA hub.
R ² mean 0.737; peak 0.797 MLP = ridge: 0.9183 = 0.9183 (gain +0.0000)	R@1/5/10 = 88/95/97% of native ceiling	ConvNeXt (held out): 96.7%

EVIDENCE LADDER — stronger claims require stronger tests

level	test	result	reading
1. Shared neighbours	mutual k-NN	strong locally	models agree most on local neighbourhoods
2. Shared order	distance-rank p	0.290 cross-modal	item rankings correlate without any map
3. Shared structure	CKA / RSA	partial	kernel alignment, sensitive to anisotropy
4. Shared subspace	ridge + retrieval	88–97% of ceiling	88.4% R@1 to 97.2% R@10 (Exp B)
5. Same geometry	rotation only	insufficient: 0.038	Procrustes R ² 0.038 vs ridge 0.592

HEADLINE EVIDENCE

0.592

Exp B adapter R²
(MobileCLIP→SigLIP)

0.771

bge–SBERT shape ρ (full 9,533
items)

96.7%

ConvNeXt hub transfer / native
(held out)

103.6%

hub vs bespoke maps (42/42
encoder pairs)

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. This project tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages. Limits: no universal or global identity; scaling is not monotonic beyond the tested range (E.16 FALSIFIED); the source-rank cliff is numerical (E.8).

ENGINEERING RESULT

From one adapter to the Cheap Adapter Hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

HUB ARCHITECTURE

Hub members (7): DINOv2-small (768-d) + DINOv2-base (1536-d) + DINOv2-large (2048-d) + GPT-2 (768-d) + BERT (768-d) + SBERT (768-d) + bge-m3 (1024-d). Concatenated, col-std scaled, whitened-PCA to 512-d.

Frozen caption head: a ridge matrix (512 → 1024) trained **once** on DINOv2-small's hub coordinates, predicting **bge-m3 text embeddings**. Frozen for all other encoders — they never train the head.

Evaluation: R@1 is scored as nearest-neighbour retrieval of the correct caption in **bge-m3's text gallery** on 1,000 held-out items.

bge-m3's dual role: it is both a hub *member* (its features help define the coordinate system) and the retrieval *target* (the head maps hub → bge text, and R@1 is scored against bge captions). This is a design choice, not a confound: bge's text embeddings are the evaluation criterion, not a free parameter.

Adding a new encoder: fit one ridge entry map (closed-form, seconds). The hub and head stay frozen. N maps, not N(N–1).

MEASURED TRANSFER — held-out bge-m3 retrieval

encoder	transfer R@1	% of native	status
DINOv2-small	0.386	98.2%	hub member; head trained here
DINOv2-base	0.437	94.8%	hub member
DINOv2-large	0.453	93.4%	hub member
SigLIP 2	—	94.2%	cross-lineage ViT, independent
ConvNeXt-base	—	96.7%	held-out CNN, ABOVE the ViT band
Random-map control	0.001	chance	negative control

42/42 encoder pairs share the hub at a mean of 103.6% of bespoke pairwise maps.

THREE COMPLEMENTARY RESULTS

MAP-FREE

$\rho = 0.290$

Mean cross-modal distance-rank agreement before any map; bge–SBERT reaches 0.771.

COMPOSED

103.6%

Mean hub performance vs bespoke pairwise maps across all 42 ordered encoder pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP adapter reaches 95.3% of native R@5. MLP ties ridge at 0.9183 (Exp B).

WHY THE INFERENCE IS CREDIBLE

Frozen models — only closed-form maps fitted

Held-out scoring — 8,533 train / 1,000 test

Negative controls — shuffle and random at chance

Nonlinear control — MLP ties linear at 0.9183 (Exp B)

PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local — nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: why shared reality matters

FractalDB tests a key PRH boundary: what structured training contributes without natural photographs, compared with matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8 — Visible geometry	G9 — Training creates geometry	G10 — Geometry is not enough
Four map-free metrics reproduce nonzero cross-modal structure. Raw shape p = 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.	Fractal training produces small but real shared geometry above matched random twins. Whitening reveals masked CNN structure, and the CNN signal survives PC1 removal.	That visible geometry does not become usable semantic retrieval through the frozen Hub: fractal models remain at the random floor, while ImageNet twins transfer strongly.

G9 — VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R (0 = random architecture; 1 = ImageNet twin): Fractal CNN 0.09 · Fractal ViT 0.21.

Matched-norm CNN control: 0.067 fractal vs 0.036 random (FractalDB norm). Norm accounts for ~10%; ~90% is genuine training. P2-cnn survives, corrected.

Robustness: CNN image-vs-text R gap 95% CI [-0.106, +0.100]; ViT [-0.053, +0.135] — both INCONCLUSIVE. Corrected drop-1: CNN ordering survives; ViT falls below random. CNN-vs-ViT contrast not interpreted (capacity differs).

G10 — FROZEN HUB/HEAD RETRIEVAL (R@1; floor = 0.050)

encoder	transfer	native	random twin	interpretation
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	—	clear functional transfer
ImageNet ViT	0.204	0.208	—	clear functional transfer

PROJECT VERDICT LEDGER — 34 pre-registered prediction units

21	7	4	2
CONFIRMED	FALSIFIED	INCONCLUSIVE	NOT MEANINGFUL
Learned shared geometry, cross-modal and cross-lineage transfer, fractal geometry above random	Fractal geometry recoverable through hub (it was not); naive scaling beyond 304M; MLP > ridge	Whether natural-world dependence differs between image and text encoders (P3 CIs straddle zero)	H3 CNN and ViT %-of-native ratios are blocked by the 0.050 floor guard; both transfer and native at floor

WHAT THE FRACTAL CONTROL ADDS TO PRH

Fractal training creates shared geometry above random, so convergence is learned rather than purely architectural. But retrieval remains at the random floor while ImageNet twins transfer strongly: shared natural-world experience is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.